QoS Enhancements for V2X Services in 5G Networks

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***Abstract* — 5G systems (5GS) have been conceived to provide, besides traditional mobile broadband services, also connectivity required by vertical industries, among which Vehicular to Everything (V2X) communication is a prominent example. Enhanced V2X communication extends its reach to safety critical services, which impose severe Quality of Service (QoS) requirements going beyond the capabilities of the early version of 5GS, corresponding to 3GPP release 15 standard specifications. This paper discusses a novel Multi Level QoS (MLQ) feature allowing a tighter QoS monitoring and control, aiming at improving service availability and continuity, essential for safety critical applications. Starting from a review of 3GPP Release 15 QoS model, the paper provides design details for the introduction of MLQ feature in the standard specifications. The paper also provides experimental evaluation of MLQ performance, combining on-field measurements and system level simulations. Results, albeit preliminary, hint to up to two-digit gain in terms of service availability and continuity, at a limited cost both in terms of complexity and additional required signaling.**

Keywords — V2X, 5G, QoS, Multi Level QoS, 3GPP

I. INTRODUCTION

Marking a discontinuity to earlier generations, 5G systems (5GS) have been conceived to satisfy requirements not only relating to mobile broadband services, but also coming from industrial applications. The 5G support of ultra-reliable communications (URLLC) and massive machine type communications (mMTC), paves the way for telecom networks to extend their range of application and to enable new business models. This represents an interesting opportunity to boost a technology sector, mobile communications, which has been stagnating over the past two decades. The willingness to seize this opportunity explains the blossoming of multiple industrial fora and organizations investigating the application of 5GS to Verticals, which has been witnessed over the past few years.

In this context, vehicular communications represent a prominent opportunity, and this is reflected on the high attention standardization bodies are paying to the investigation of Vehicular to Everything (V2X) use cases [1], the definition of their requirements [2], and the design of relating solutions [3], [4]. Although 4G networks already provided the support of basic V2X services and some further enhancements had been introduced in the earliest version of 5GS (corresponding to 3GPP release 15 standard specifications), the emphasis recently given to safety critical applications combined with the relating updated requirements highlighted the necessity of new solutions, in particular in the domain of Quality of Service (QoS).

This paper presents the novel Multi Level QoS (MLQ) feature, an enhancement for 5GS QoS to support safety critical V2X services over Uu interface. The solution, proposed as an architecture enhancement for 3GPP release 16 standard specifications, leverages on 5G capability to interact with 3rd party's application functions (AFs), and aims at providing the network with the ability to smoothly adjust the QoS of sessions according to network performance dynamicity, yet taking into account the QoS levels at which V2X applications can be supported. The MLQ feature aims at maximizing service quality and avoiding service interruption.

The paper is structured as follows: an overview of the updated V2X service requirements for 5GS, included in Section II, provides the motivation of this work. Section III briefly reviews the 5G QoS model, emphasizing its limitations, and illustrates the proposed MLQ solution. Section IV contains a preliminary performance evaluation of the proposal, and Section V concludes the paper.

II. ENHANCED V2X (EV2X) REQUIREMENTS FOR 5G

V2X service requirements definition dates back in time and their analysis has been an uninterrupted running process since then. Recent updates in this respect highlighted new perspectives.

A first aspect to note is the progressively higher relevance given to safety critical applications ([1]), this leading on one hand to more challenging performance requirements and, on the other hand, to the need of a stricter reliability in the requirements fulfillment. For example [2], vehicle platooning services require end to end latency down to 10ms with 99.99% reliability, while advanced driving services may require end to end latency as little as 3ms and bitrate up to 50Mbps with 99.999% reliability.

A second important aspect is documented in [2]: advanced V2X applications can be provided with different Levels of Automation (LoA). The LoA of a V2X application defines [2] "the necessary degree of human interaction with the application and the role of the automated system in monitoring and controlling the vehicle driving environment". Five different LoAs have been defined ([2]): No Automation, Driver Assistance, Partial Automation, Conditional Automation, High Automation and Full Automation. Although intuitive, it should be noted that for a given V2X application, a higher LoA imposes stricter requirements to the communication system in terms of QoS and reliability. For example [2], Cooperative Lane Change service can be provided with low or high LoA: for low LoA, end to end latency of 25ms with 90% reliability is required, while for high LoA the requirement changes to 10ms with 99.99% reliability.

From the two aspects briefly analyzed, two considerations can be made. First, the support of services which may have severe impact on human safety forces telecom networks to improve the accuracy and reliability of performance requirements fulfillment and QoS provisioning. Second, the possibility to provide V2X services with different LoAs, corresponding to different QoS requirements, hints to possible 5GS enhancements ensuring service continuity despite of QoS dynamicity inherently characterizing mobile networks.

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III. 5G QOS ENHANCEMENTS FOR EV2X

A. The 5G QoS Model and V2X QoS Prior Art

To understand the limitations of the earliest version of 5GS (defined by 3GPP release 15 standard specifications), 5G connectivity and QoS models are briefly illustrated in this section. 5G connectivity is built upon the Packet Data Unit (PDU) Session concept [5]: a logical association between a UE and an external Data Network (DN) through which 5G network provides bidirectional User Plane (UP) data transfer. The PDU Session is established by 5G Core network (5GC) and it identifies all UP entities and interfaces involved in the data transfer: the UE, the Access Node, the Uu radio interface between UE and Access Node, the User Plane Function UPF, the N3 interface between Access Node and UPF, and the N6 interface between UPF and external DN. The establishment, modification and release of PDU Sessions are handled by the Session Management Function (SMF) according to network policies controlled by the Policy Control Function (PCF), both Control Plane (CP) functions of the 5GC.

Each PDU Session can consist of multiple QoS Flows. A QoS Flow defines the finest granularity of QoS differentiation within a PDU Session. Without entering in complex details unnecessary to this dissertation, it has to be stressed a QoS Flow is characterized by a QoS Profile, which fully specifies the quality of service the flow will receive. To this end, a QoS Profile includes a set of parameters defining e.g. the Packet Delay Budget (ms), the Packet Error Rate (%); the Guaranteed and Maximum Bitrate (kbps) for the QoS Flow.

When a UE needs a service provided by a DN which requires data exchange between UE and DN with a given QoS, the 5GS provides connectivity by establishing a PDU Session between the UE and DN and, within the PDU Session, it sets up a QoS Flow according to the service QoS requirements.

At PDU Session establishment, QoS Flows are set up by the SMF. The QoS Profile and related parameters for the QoS Flow are defined according to network policy set into PCF. During establishment, the SMF communicates the QoS Profiles for the QoS Flows to the Access Node, the UE and the UPF, which will enforce the corresponding QoS via specific radio and core network resource management algorithms (e.g. resource reservation, scheduling etc.).

It may of course happen some of the elements involved in the data transfer (most probably the Access Node) are not able to ensure the fulfillment of the QoS Profile for all QoS Flows during the whole PDU Session lifetime. This, at Access Node, may be due to a variety of reasons, including access network congestion, poor quality of the radio connection, excessive data generated for the QoS Flows. To cope with such situations, 5G defines the Notification Control mechanism: whenever the Access Node cannot fulfill the QoS Profile for a QoS Flow, it starts a best effort handling of the flow and sends a notification to SMF. The SMF forwards the notification to the PCF, and 5GC may initiate a modification or release of the QoS Flow.

Drawbacks of this model for eV2X services are straightforward. Following the inability to fulfill QoS Profile for the QoS Flow associated to the eV2X service, due to the best effort handling, the service may incur in QoS downgrade not compatible with its performance requirements. This may likely result in unexpected service interruption (unexpected as the V2X application layer is not notified in any way on the QoS status, and will just experience an unacceptable QoS downgrade leading to service termination). Service interruption would also follow a QoS Flow release which 5GC could trigger. Even in case a QoS Flow modification takes place, it is unclear how the modified QoS Flow could possibly match with service QoS requirements. Hence, also in this case service termination may be expected. Although strongly depending on implementation, it is reasonable to assume the eV2X application to attempt a reestablishment of the PDU Session following the service termination.

A significant amount of research results on QoS enhancements for V2X can be found in the literature. In [6], adaptive numerology for 5G physical layer is proposed to increase QoS reliability against the effect of mobility and dynamic channel conditions. Still focusing on radio enhancements to improve QoS reliability, but shifting the attention to higher layers, [7] proposes a multi-RATs scheme exploiting diversity by combining transmission over LTE-Uu and PC5 interfaces. A similar approach is also discussed in [8], where strategies for multi-connectivity management are evaluated. Keeping the focus on higher layers, [9] proposes the implementation of an enhanced MAC layer, featuring a novel channel access method for QoS provisioning in high mobility scenarios. Other recent works focus on QoS improvements for V2X device to device communications, e.g. [10] where the authors present a low-cost cluster-based method featuring a cluster members' data estimation mechanism to improve V2X QoS. Finally, some works investigate the application of machine learning techniques for QoS improvements. For example, [11] illustrates a supervised learning based QoS assurance architecture for 5G networks enabling fast reaction to QoS degradation and high accuracy detection of the root causes of QoS anomalies.

To the best of authors' knowledge, the vast available literature, herein represented by a significant albeit limited sample, includes an uncountable list of solutions improving different aspects of QoS, but fails to provide system mechanisms handling the not so infrequent real scenarios where target QoS cannot be fulfilled. The MLQ solution enables controlled QoS adjustments, triggered by the dynamic radio conditions, towards QoS levels yet compatible with alternative requirements, and allows a timely adaptation of the eV2X application.

B. 5G QoS Model Enhacements: Multi Level QoS

The Multi Level QoS (MLQ) solution has been designed to enhance 5G QoS model and overcome the manifest shortcomings described in Sec. III. A.

The solution exploits a prominent 5G feature: the capability of 5GC to interact with 3rd parties' Application Functions (AFs). A 3rd parties' AF is an external Network Function (NF), not belonging to the mobile network operator administrative domain. In 5GS, AFs can become integral part of the 5GC CP via a Network Exposure Capability function NEF, which regulates AFs interaction with 5GC NFs. Third Party's AF is, in this context, a software function developed by the eV2X service provider.

The high level functional description of MLQ is illustrated in Fig. 1. The solution defines a new interaction mechanism between AF and PCF (via NEF), allowing AF to indicate, for a given V2X service, the set of QoS Levels at which the service can be provided (step 1).

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The set of QoS Levels includes the preferred level (i.e. the target QoS) as well as alternative levels (i.e. back-up options). Each QoS Level is defined in terms of V2X service requirements, e.g. required bitrate, maximum packet delay, packet error rate, communication reliability etc. Upon the AF indication of the multiple QoS Levels for a V2X service, the PCF translates the QoS Levels into QoS Profiles, and creates a Multi Level QoS policy to apply to the V2X service (step 2). The policy specifies that QoS Flows set up for PDU Sessions relating to the V2X service are associated not to a single QoS Profile but, rather, to the set of QoS Profiles derived from the QoS Levels. Such flows are denoted as Multi-QoS Profile Flows.

When a UE requests 5G connectivity for the V2X service, during the PDU Session establishment (step 3), the needed QoS Flows are set up. In this phase, the PCF applies the Multi Level QoS policy, commanding the SMF to set up Multi QoS Profiles Flows, transferring to SMF the profiles derived by the multiple QoS Levels (step 3.1). It should be noted the SMF sets up the Multi QoS Profile Flows interacting with all UP entities: Access Node, UPF and, indirectly, UE (steps 3.2 and 3.3). For all UP entities and interfaces, the availability of resources to fulfill the preferred QoS Profile is verified. If UP resources are sufficient, the flow will provide the preferred QoS. Otherwise, UP resource availability for alternative QoS profiles would be checked and, eventually, an alternative QoS profile provided.

After the successful flow set up, the V2X service data transfer can take place (step 4). During the PDU Session lifetime, if the Access Node cannot fulfill the QoS Profile provided, it verifies if any alternative QoS Profiles can be fulfilled, and notifies the QoS Profile downgrade to the SMF, in compliance to the

Notification Control mechanisms (steps 5 and 6). The change in the provided QoS will be instructed to all UP entities (step 7). In other words, 5GC enforces a smooth adjustment of the QoS Profile of the Multi QoS Profile Flow, according to Access Node capabilities. The smooth adjustment ensures V2X service continuity, as all QoS Profiles are compatible with the QoS Levels acceptable by the V2X service. Finally (steps 8.1 and 8.2), to allow a prompt reaction of the V2X application (step 9), a QoS Level change notification is sent both to AF (via PCF) and UE (relayed by AN).

MLQ does not only enable smooth QoS downgrade aiming at avoiding service interruption. In fact, the feature allows also upgrades of the QoS Profile of the Multi QoS Flow, aiming at restoring whenever possible the preferred QoS Profile, hence maximizing the QoS provisioning.

Finally, it is relevant to note the MLQ feature has been designed also targeting improvements in the QoS provisioning during UE mobility handling. During the handover preparation procedure, by sending the HO Request message, the source AN provides the target AN with the list of QoS Flows to handover. For Multi QoS Profile Flows both the preferred QoS Profile and the alternative QoS Profiles will be provided. Intuitively, the Multi QoS Profile Flows allow for higher flexibility during the admission control in the target AN: if resources are insufficient to admit a Multi QoS Flow according to its preferred QoS Profile, the flow might still be handed over successfully, if available resources are compatible with any alternative QoS Profile requirements.

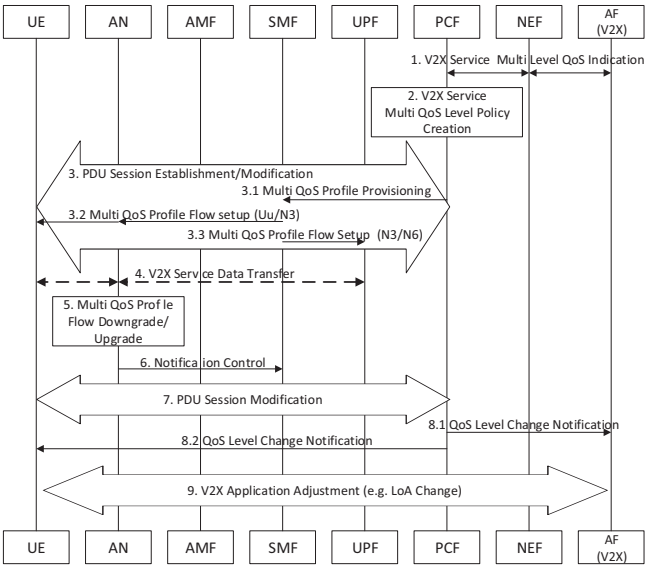


Fig. 1. Multi Level QoS Feature (High Level Functional Description)

IV. Performance Evaluation

An accurate engineering of the MLQ feature requires experimental activity to understand benefits, costs and possible drawbacks which may manifest due to the complexity of real deployment scenarios. To this end, the effects of the feature introduction in a deployed 4G network were experimented. With full awareness of the only partial reach of the results, the experimentation on a 4G network was justified by two reasons: first, large scale 5G deployments are currently unavailable and, second, although conceived for 5GS, the extension of MLQ to legacy (i.e. 4G) systems might also be considered.

A. Experimental Set Up

In the first step of the experimentation, the coverage map of a commercially deployed 4G network was created. An urban area of 60E4 square meter was considered, covered by eleven 20MHz LTE cells at 1.8GHz and 2.1GHz. The map was created upon network KPIs collected via drive tests, performed over all roads accessible by vehicles of the considered area, on tracks summing up to approx. 4.5km. To increase reliability, drive tests were repeated multiple times for a total measurement time of approx. 15h. Measurements, collected with a spatial granularity of 10m, included Timestamp, Longitude, Latitude, Altitude, Cell ID, as well as Rx Level (RXLev), Rx Quality (RXQual), SNR, RSSI, ARFCN. The collected measurements allowed an accurate statistical description of network KPIs with a resolution of approximately 25x25(sqm).

In a second step, the network coverage map was embedded in a 4G system level software simulator. The software simulates V2X services data traffic generated by vehicles driving in the area, and the data traffic transmission through the 4G network. Extended Sensor use case was considered [2], in particular the V2X service for sensor information sharing between vehicles. Three QoS Levels were defined, corresponding to the three QoS requirements set indicated in Table I. Network performance with / without the MLQ feature were investigated. In the cases the MLQ feature is disabled, the network is assumed operating with the sole knowledge of QoS Level #1.

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TABLE I. EXTENDED SENSORS: QOS LEVELS AND REQUIREMENTS

QoS Level	QoS Requirements		
	Max E2E Latency (ms)	Reliability (%)	Data Rate (Mbps)
#1 (Preferred)	3	99.999	5
#2 (Alternative)	10	99.99	2.5
#3 (Alternative)	50	99	1

B. *Simulation Parameters Tuning*

A preliminary analysis of the network KPIs collected for the network coverage map highlighted an average system capacity for the whole 11-cell network of approximately 1.8Gbps (estimated by statistical elaboration of network coverage KPIs over the whole area accessible by vehicles). Considering the system parameters setting indicated in Table II (4x4 MIMO transmission, highest modulation 64QAM) a maximum theoretical system capacity of about 3.3Gbps can be computed. The significant reduction with respect to the maximum capacity can be attributed to the non-ideal network deployment characterizing the real urban scenario, where e.g. several narrow roads may experience poor coverage, which forces the use of lower modulations and coding schemes and hence leads to low peak data rate. A quantitative evidence is immediate just examining the Channel Quality Indicator (CQI) statistical distribution over the whole 60E4 square meter area, shown in Fig. 2. 4G system provides the peak capacity only for CQI equal to 14 and 15, while a capacity reduction exceeding 50% is already experienced for CQI equal to 9.

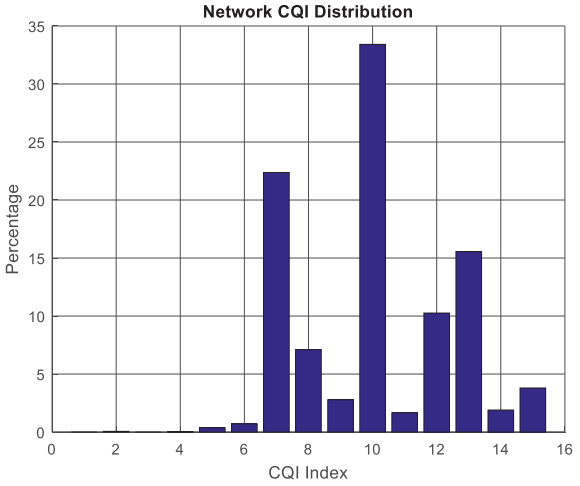


Fig. 2. CQI (Channel Quality Indicator) distribution over the test area.

In addition, preliminary simulations (run for simulator tuning purpose) have shown that a V2X traffic with a uniform geographical distribution over the roads of the test area does not imply a uniform distribution of load among the 11 cells. Rather, 75% of the load is absorbed by only 3 cells. This apparent inconsistency actually has a reasonable explanation. Although drive tests showed 4G coverage in the area was provided by 11 cells, such cells have been actually deployed to cover a much wider area. This emerges clearly examining the cumulative distribution function of the measured RXLev for the 11 cells, depicted in Fig. 3.

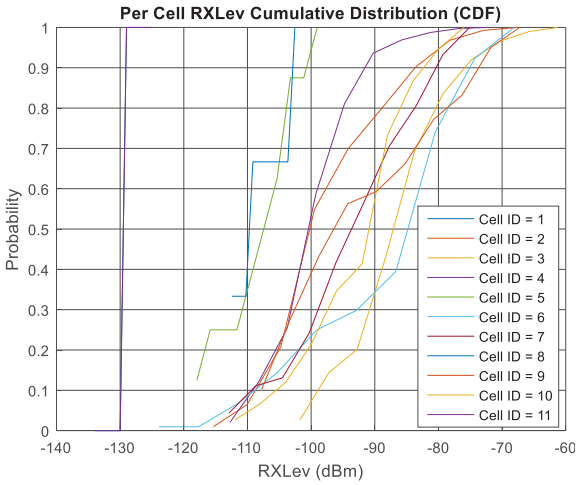


Fig. 3. Per Cell RXLev distributions over the test area.

For several cells (e.g. corresponding to Cell ID equal to 5, 8 and 11) an RXLev always lower than -100dBm and with high probability lower than -110dBm was measured. Clearly, those cells have been deployed to cover neighbor areas, adjacent to roads where drive tests were run.

This analysis drove the tuning V2X traffic simulation parameters. V2X service sessions lasting 5min with up to 5Mbps required data rate were considered. Session generation rates ranging between 0.05 and 0.5 (1/s) have been considered, this resulting in an average traffic load ranging between 75 and 750Mbps. Table II summarizes the simulation parameters setting.

C. *Simulation Results*

As discussed in Sec. II and Sec. III, the MLQ feature aims at enhancing the support of safety critical V2X services, for which service availability should be maximized and service interruption minimized. For this reason, improvements determined by the feature introduction have been evaluated observing three critical KPIs: the Session Rejection Rate (i.e. the ratio between the number of failed session establishments and the total number of session establishment attempts), the Handover (HO) Failure Rate (i.e. the ratio between the number of failed HO due to resource unavailability in the target cell and the total number of HO attempts), and the Session Drop Rate (the ratio between the number of sessions dropped and the total number of sessions simulated). Fig. 4 plots Session Rejection Rate and Handover Failure Rate against the generated load.

TABLE II. SIMULATION PARAMTERS

Parameter	Values
Cell Bandwidh	20MHz (200PRBs)
Cell Band	1.8 – 2.1 GHz
Radio Configuration	MIMO 4x4 (Q-PSK - 64QAM)
V2X Service Generation Rate	0.05 – 0.5 (1/s)
V2X Service Duration	300s
Simulated Time	5h

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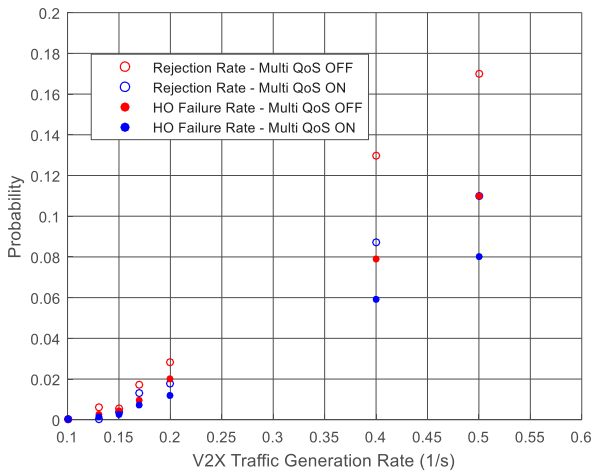


Fig. 4. Session Rejection Rate and Handover Failure Rate vs. Load.

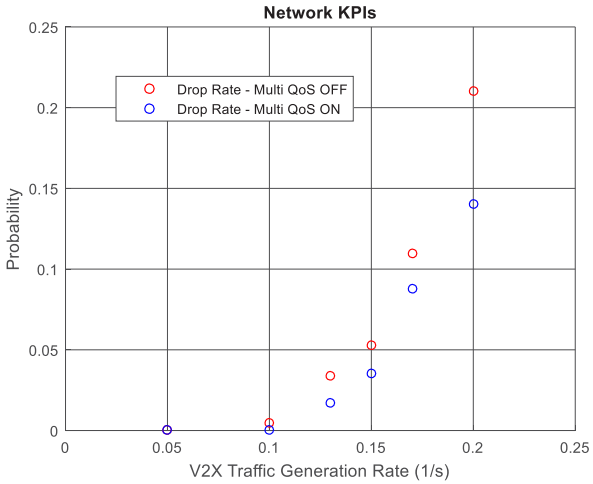


Fig. 5. Session Drop Rate vs. Load.

A first necessary observation is due to the relatively high Session Rejection Rate experienced (ranging between 1% and 3%) even at very moderate session generation rates of 0.15 – 0.2 (1/s), corresponding to system loads of 225 – 300 (Mbps). These high values can be attributed to the unbalanced distribution of traffic among the cells, which reasons has been explained in Sec. IV. B. Focusing on the Session Rejection Rate, enabling MLQ results in a very significant reduction of the rejection rate, up to 36%. A reduction of the same order of magnitude can be observed on the Handover Failure rate. Fig. 5 plots Session Drop Rate against the generated load. The very high values in absolute terms shown in the chart (5% – 20% for loads of 225Mbps – 300Mbps) can be attributed to two combined effects. Once again, the unbalanced distribution of traffic determines load situations close to congestion for a few cells. At the same time, device mobility leads to a quite significant variability of the radio channel quality, which implies a significant variability in the required radio resources to satisfy the QoS profile. For example, considering the two most probable CQI values 7 and 10 (as shown in Fig. 2), the required number of resource elements (REs) to provide 5Mbps is equal to 899 and 450 respectively. The two combined effects make it probable for highly loaded cells to drop sessions when their radio resource demand sharply increases as the device moves. That explained, Fig. 5 shown an up to 2-digit reduction of Drop Rate thanks to the introduction of the MLQ feature.

V. Conclusions

The relevance of vehicular communications as field of application for 5GS triggered intense activities aiming at the definition of network enhancements fulfilling challenging V2X service requirements. In this context, this paper presented the novel Multi Level QoS (MLQ) feature, candidate enhancement for 3GPP release 16 standard specifications. MLQ feature aims at improving service availability and continuity, specifically targeting safety critical V2X services. The feature has been designed leveraging on the 5GS capability to interact with 3rd parties Application Functions (AFs) and exploiting the Level of Automation (LoA) concept, which enables the possibility to support V2X services with different levels of QoS.

The MLQ feature defines a mechanism making 5GS aware of the alternative QoS levels a V2X service can be provided, and enables smooth transitions amongst the QoS levels according to the actual radio link quality and radio resource availability. The benefits of the MLQ feature have been evaluated via an experimental activity, which highlighted 2-digit gain in terms of Session Rejection Rate, Session Drop Rate and Handover Failure Rate.

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